

Heart Disease Prediction Report



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Internship Project 3

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Project Synopsis

This project classified individuals as having or not having heart disease using demographic (Age, sex, ethnicity, socioeconomic indicators), clinical (Blood pressure (systolic/diastolic), cholesterol, glucose, family history among others, and lifestyle (Smoking status, alcohol use, physical activity levels, BM etc.) attributes. It developed a robust, well-calibrated classifier, understanding which features drive predictions, and ensuring the model is fair, reliable, and clinically meaningful after efficient data cleaning processes. Modelling approach used were Logistic regression, Random Forest and XGBoost. They were subjected to training and evaluation using algorithms with metrics such as accuracy, precision, recall, F1-score. The findings indicated Gradient Boosting as the best predictor. It captured complex clinical and lifestyle patterns. Random Forest model was a strong alternative, while Logistic Regression model was useful for baseline.

Problem Statements

- Heart disease remains one of the leading causes of death globally, placing a heavy burden on healthcare systems and families.
- Many cases go undetected until late stages, when treatment options are limited.
- Traditional diagnostic methods can be costly, invasive, or dependent on specialist interpretation.
- There is a pressing need for early detection tools that can identify individuals at risk using readily available medical and lifestyle data.

PROJECT

Goal

Build a predictive, interpretable machine learning model that leverages medical and lifestyle data to improve early detection and patient outcomes.

Objectives

The objectives of this project is to develop a machine learning model that:

- ❑ Predicts the presence or absence of heart disease using patient attributes such as cholesterol, blood pressure, glucose, BMI, and lifestyle factors.
- ❑ Analyzes patterns across demographic, clinical, and behavioral variables to uncover risk indicators through structured dataset.
- ❑ Supports preventive care by enabling timely intervention and reducing mortality through early diagnosis.

Dataset Overview

- ◆ Total Records: 70,000 patients
- ◆ Collected from multiple clinical sources, covering demographics, lifestyle habits and clinical measures such as blood pressure, cholesterol, glucose and BMI.
- ◆ Target variable: Presence or absence of heart disease

Data Cleaning and Preprocessing

- ◆ Parsed mixed-format dates and extracted temporal features
- ◆ Converted age from days to years
- ◆ Corrected blood pressure anomalies and outliers
- ◆ Handled missing values using median and mode imputation
- ◆ Scaled numeric features, encoded categorical variables and binary flags inputed.

Feature Engineering

- ◆ Derived BMI from height and weight,
- ◆ Calculate pulse pressure, and mean arterial pressure (MAP)
- ◆ Created hypertension stage flags (Stage 1, State 2, Any) and binary lifestyle indicators
- ◆ Binned age and BMI into categorical ranges
- ◆ Encoded categorical variables using one-hot encoding
- ◆ Added temporal features like year, month, and day of week enriched the dataset

Modeling Methodology

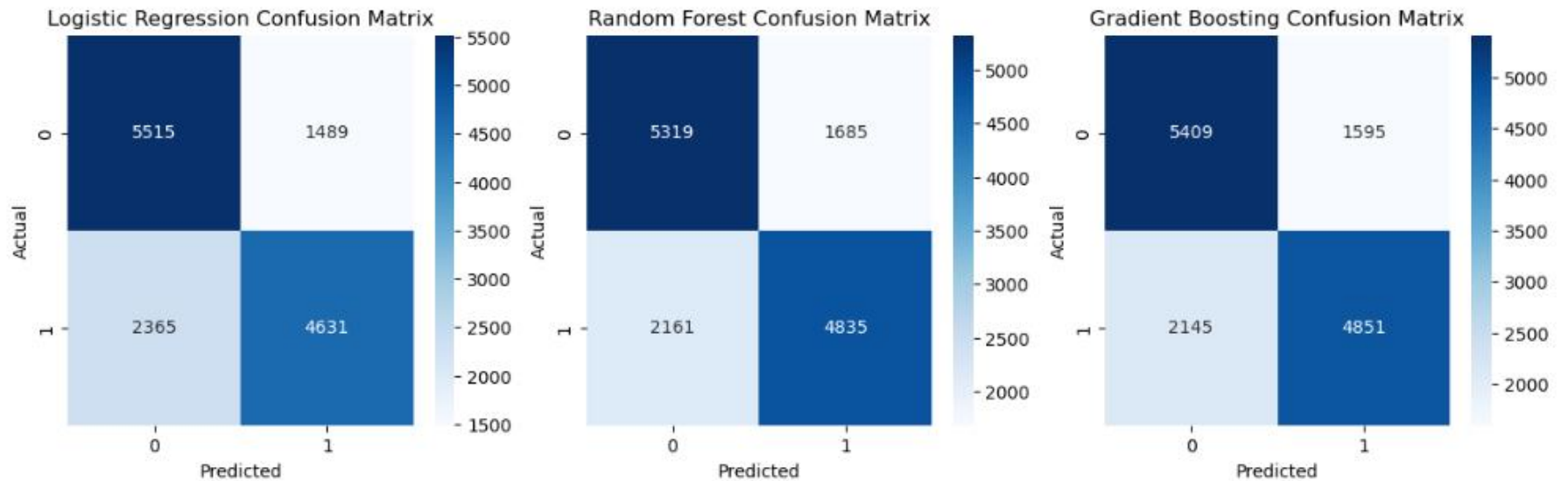
Three models trained and evaluated:

- ❑ Logistic Regression: Used for baseline, interpretability (with class balancing)
- ❑ Random Forest: Used for robust, high accuracy (balanced subsample)
- ❑ Gradient Boosting: For performance (default parameters)
- ❑ Evaluation was based on metrics such as accuracy, precision, recall, F1-score, and ROC-AUC with 5-fold Stratified Cross-Validation for stability

Major Findings

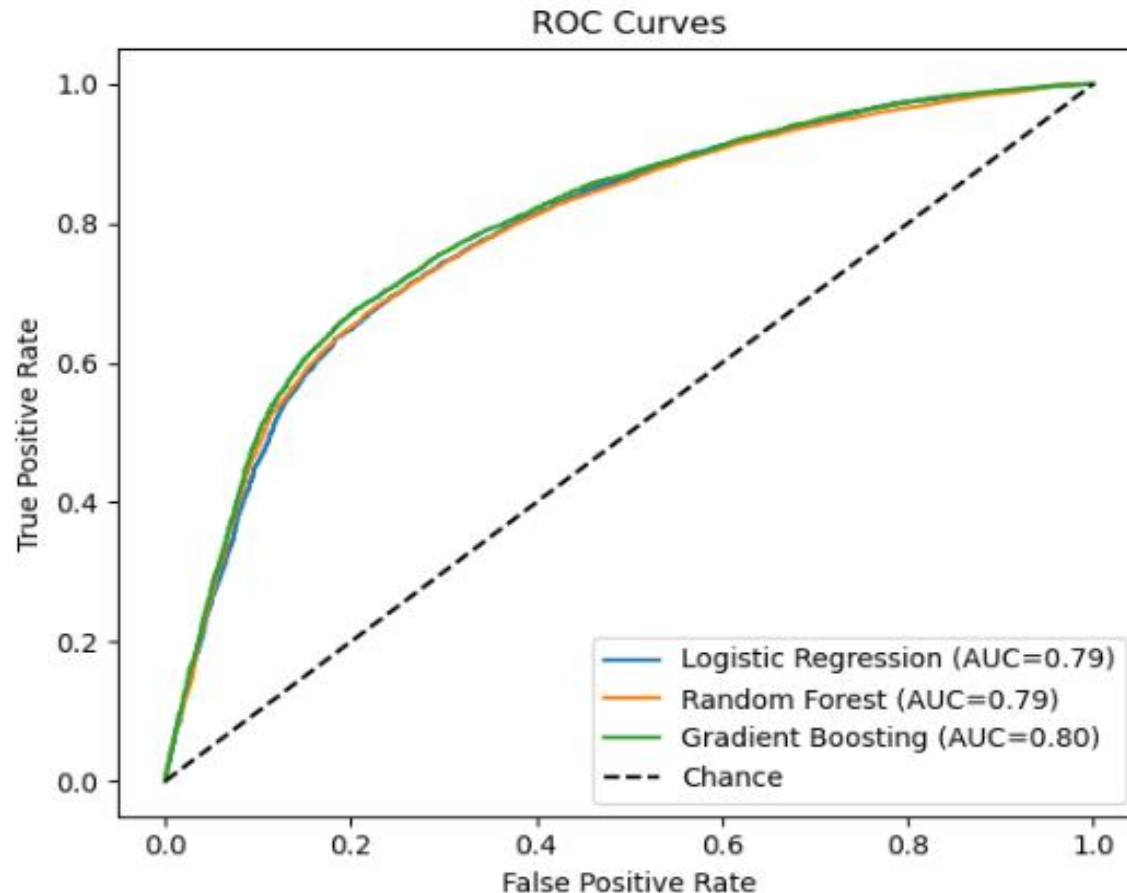
Confusion Matrices

Confusion matrices and ROC curves plotted successfully



- ❑ Logistic Regression: More false negatives, fewer false positives.
- ❑ Random Forest: Balanced sensitivity and specificity.
- ❑ Gradient Boosting: Highest true positives, lowest false negatives.

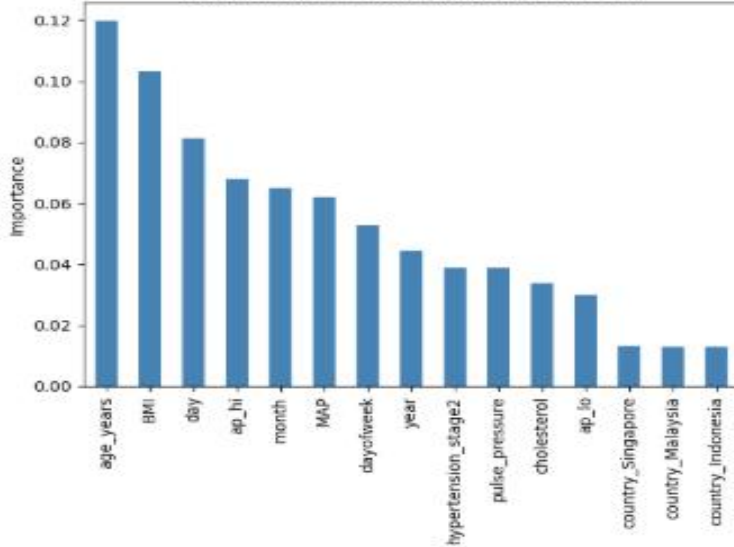
ROC Curve Comparison



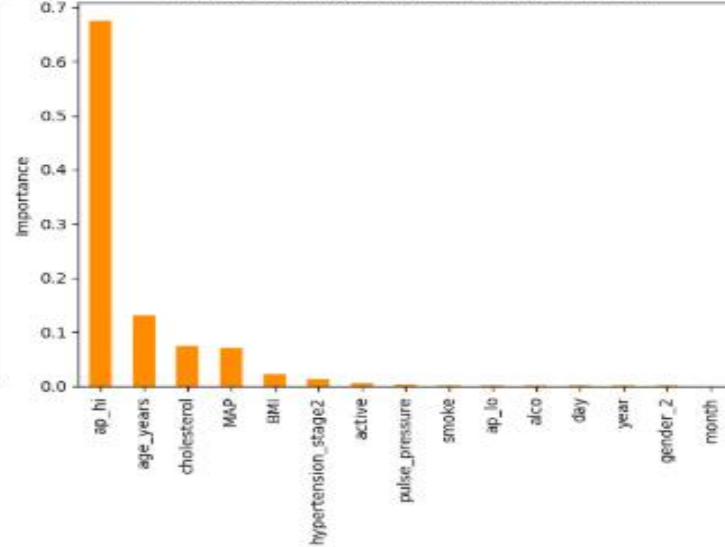
- ❖ All models outperform random guessing
- ❖ Gradient Boosting achieved highest AUC (0.80).
- ❖ Logistic Regression and Random Forest tied at AUC = 0.79.

Feature Importance

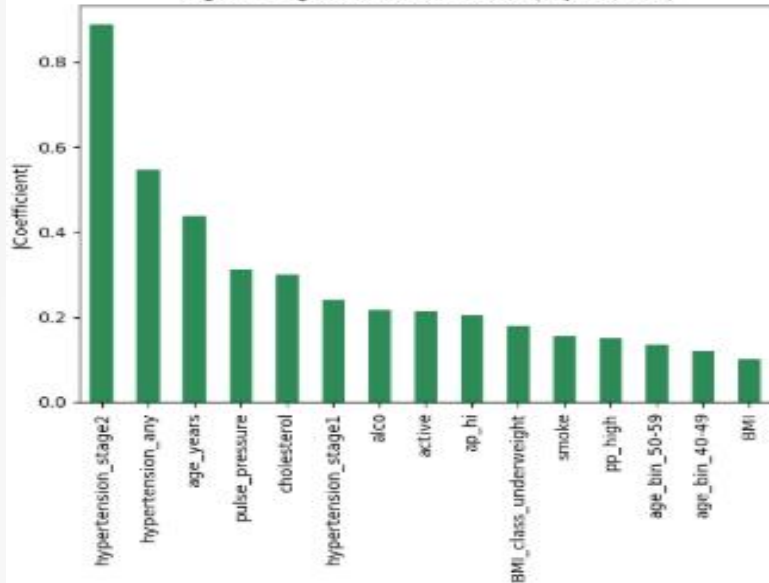
Random Forest Feature Importance (Top 15)



Gradient Boosting Feature Importance (Top 15)

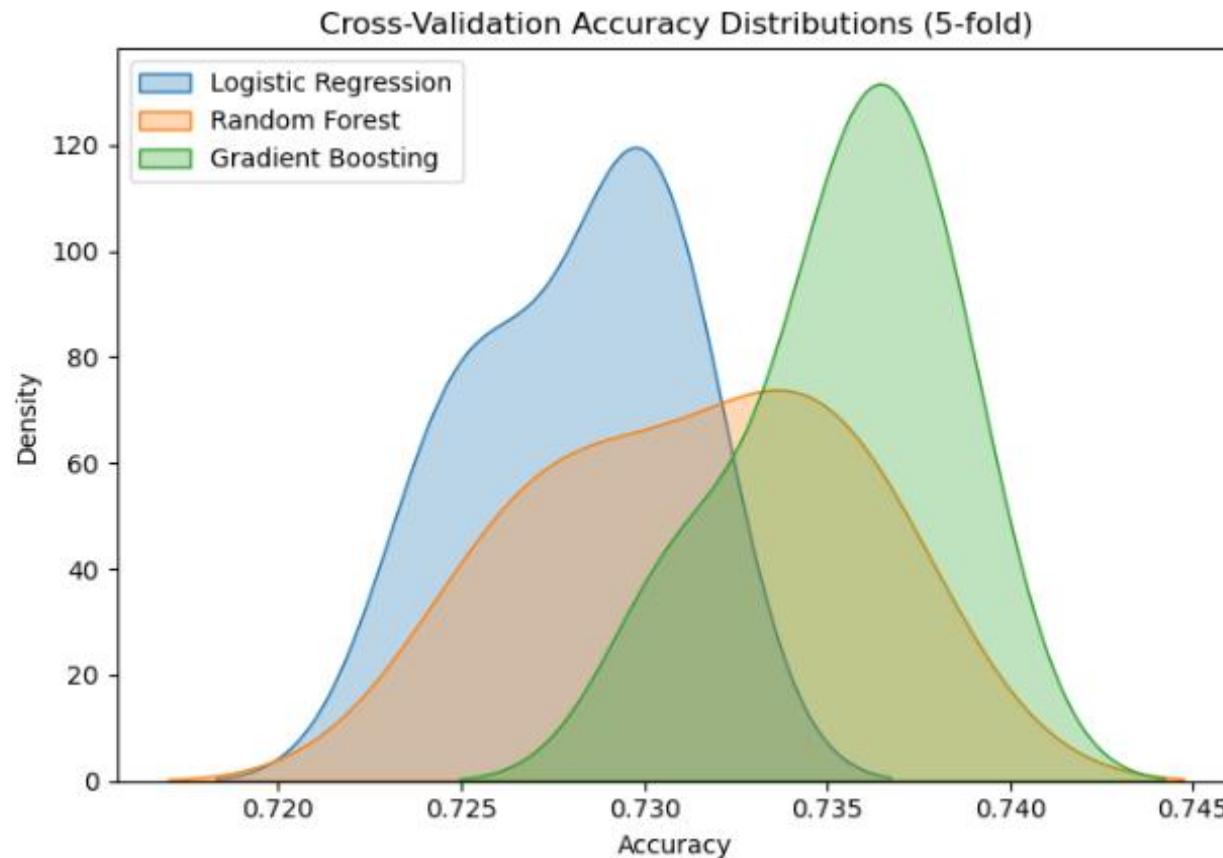


Logistic Regression Coefficients (Top 15, abs)



- ❖ Age, glucose, BMI, hypertension consistently top predictors.
- ❖ Gradient Boosting relied heavily on age.
- ❖ Logistic Regression emphasized hypertension.

Cross-Validation Accuracy Distributions



- ❖ Gradient Boosting showed most consistent accuracy across folds.
- ❖ Random Forest slightly higher mean accuracy but more variability.
- ❖ Logistic Regression lowest peak.

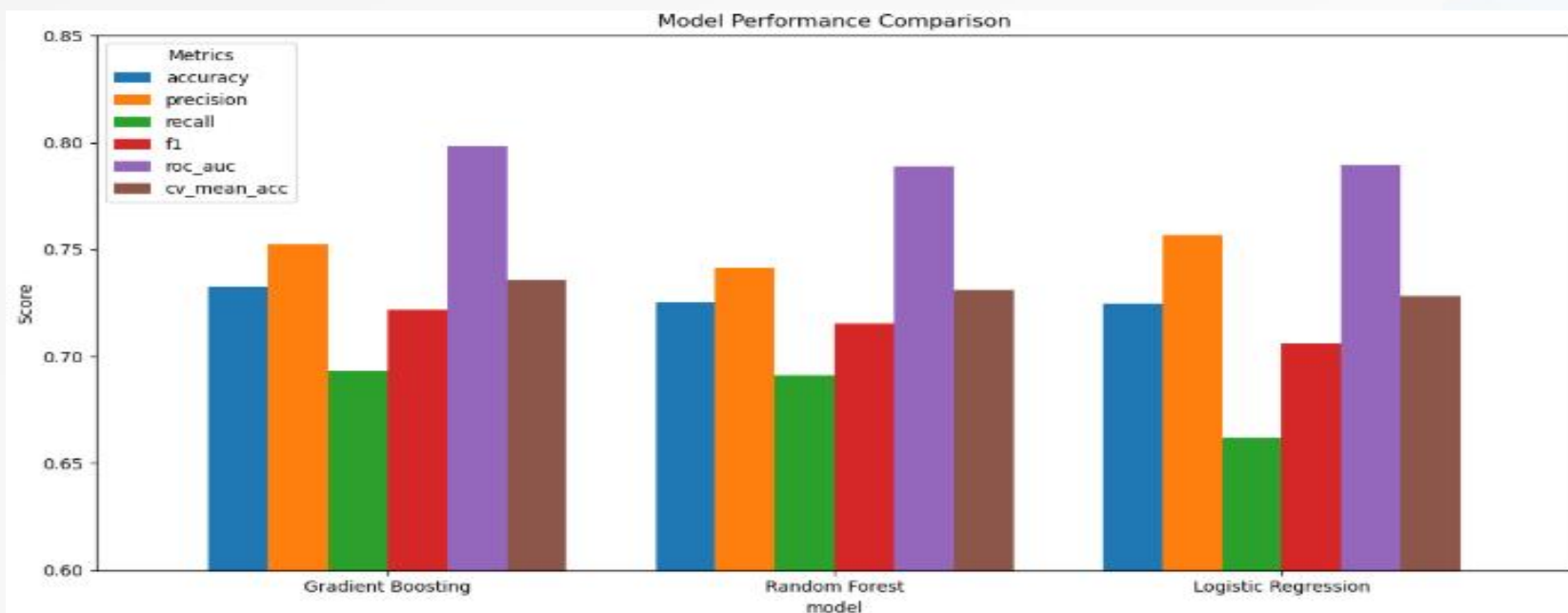
Model Performance Summary

Model Performance Summary:

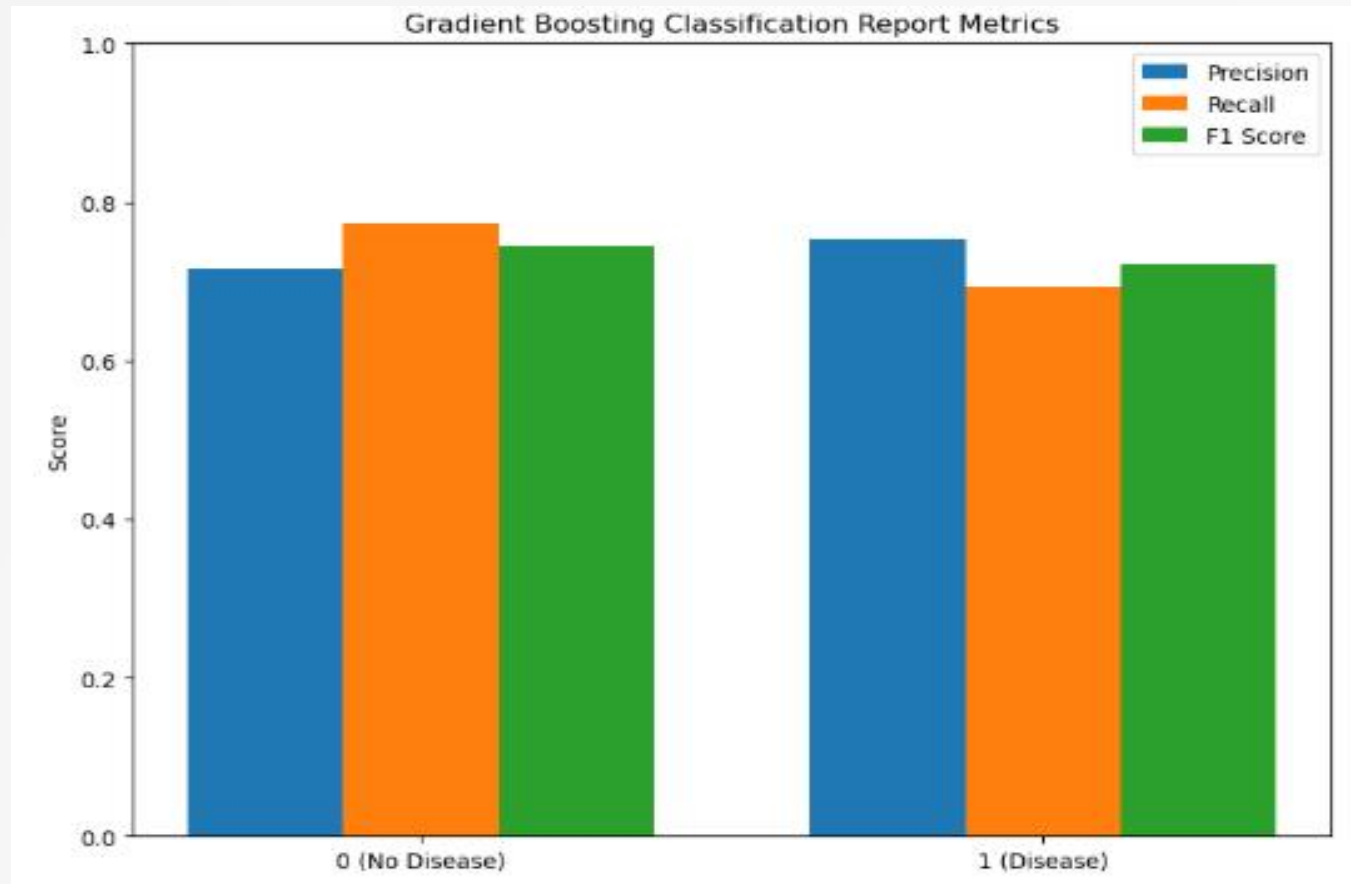
	model	accuracy	precision	recall	f1	roc_auc
2	Gradient Boosting	0.7329	0.7526	0.6934	0.7218	0.7981
1	Random Forest	0.7253	0.7416	0.6911	0.7154	0.7888
0	Logistic Regression	0.7247	0.7567	0.6619	0.7062	0.7897

	cv_mean_acc	cv_std_acc
2	0.7354	0.0025
1	0.7314	0.0037
0	0.7281	0.0025

- ❑ Gradient Boosting: Accuracy = 72.95%, F1 = 72.18%, ROC AUC = 79.81%. Best overall balance of precision, recall, and stability.
- ❑ Random Forest close second;
- ❑ Logistic Regression best for interpretability.



Gradient Boosting: Classification Report



- ❖ Class 0 (no disease): Recall = 0.77.
- ❖ Class 1 (disease): Precision = 0.75.
- ❖ Overall accuracy = 0.73.

Summary of Findings



- ◆ XGBoost is the best model
- ◆ Most accurate in classifying heart disease
- ◆ Most stable across validation folds
- ◆ Captures complex clinical + lifestyle patterns
- ◆ Random Forest: strong alternative
- ◆ Logistic Regression: useful baseline

Key Risk Factors



- ◆ Age (older patients at higher risk).
- ◆ BMI (overweight/obese categories strongly associated with disease).
- ◆ Blood Pressure (ap_hi, ap_lo, MAP, pulse pressure; hypertension stages critical predictors).
- ◆ Cholesterol (elevated levels linked to higher risk).
- ◆ Lifestyle (smoking, alcohol consumption, physical inactivity).

Preventive Measures



- ◆ Reduce alcohol intake to lower cardiovascular strain.
- ◆ Manage blood pressure through monitoring, medication, and lifestyle changes.
- ◆ Maintain healthy BMI with balanced diet and regular exercise.
- ◆ Monitor cholesterol via diet (reduce saturated fats, increase fiber) and medical management.
- ◆ Encourage physical activity (≥ 30 minutes moderate exercise most days).
- ◆ Smoking cessation to reduce cardiovascular risk.

Conclusions



- ◆ Gradient Boosting achieved highest AUC (0.80) and best F1 score.
- ◆ Most important predictors: age, glucose, BMI, hypertension
- ◆ Gradient Boosting showed stable performance across folds.
- ◆ Logistic Regression is useful for clinical communication
- ◆ Random Forest indicated robust alternative.

Recommendations



Technical

- ✓ Adopt Gradient Boosting as the primary model (best accuracy & ROC AUC).
- ✓ Perform hyperparameter tuning to optimize performance.
- ✓ Integrate SHAP explainability for global and patient-level insights.
- ✓ Expand feature engineering with interaction terms (Age $\times$ BMI, Cholesterol $\times$ Hypertension).
- ✓ Validate on external datasets for generalization.

Clinical

- ✓ Use the model as a screening tool to flag high-risk patients.
- ✓ Prioritize recall (sensitivity) to minimize missed disease cases.
- ✓ Provide clinicians with interpretable outputs (SHAP waterfall plots, feature importance).
- ✓ Highlight key risk factors (age, BMI, blood pressure, cholesterol, MAP).
- ✓ Collaborate with healthcare professionals to align predictions with real patient outcomes.

Future Work



-  **Explore deep learning approaches with larger datasets.**
-  **Investigate longitudinal analysis using temporal features.**
-  **Incorporate lifestyle interventions (smoking cessation, activity levels) into predictive recommendations.**